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Formation and Characterization of Transparent Conductive Oxide SnO_2:Ta Grown at Low Deposition Temperature

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ABSTRACT

The formation of the transparent conductive oxide SnO_2:Ta by reactive direct-current magnetron sputtering at a low temperature of 225°C was studied as a function of the oxygen flow rate using a pilot-scale setup. Beginning with a mixture of β-Sn and SnO as majority phases and minority fractions of Sn_3O_4 and SnO_2 at very low O_2 flows, up to the formation of phase-pure rutile-type SnO_2:Ta at higher flow rates, a systematic evolution of the phase structure is observed. The SnO_2:Ta (0.70 at.% Ta) film grown at optimized flow conditions exhibits a transmittance of >80% from 500 to 2500 nm and a specific resistivity of 0.3 Ω cm. While the transmittance is competitive, the resistivity is in the upper range of that reported for low-temperature-deposited SnO_2:Ta films in research lab studies. Based on X-ray diffraction, Raman spectroscopy, optical spectroscopy, and X-ray photoelectron spectroscopy, we conclude that the small crystallite size, microstrain, and point defects caused by the low growth temperature might prevent charge carrier activation, even though the Ta dopant is incorporated as Ta^{5+} in the rutile-type SnO_2 lattice.

1 | Introduction

Among the three classical transparent conductive oxides (TCOs) such as In_2O_3, ZnO, and SnO_2, the latter is distinguished by a high chemical, mechanical, and thermal stability [1–3]. The most abundantly used dopant for SnO_2 is fluorine [4–7]. SnO_2:F can be used as a transparent electrode in display technologies, as a low-emissivity glass coating to improve the energy efficiency of buildings by reducing the infrared heat transmission while still retaining the transparency for visible light, or as a spectrally selective coating for solar collectors [3, 8, 9]. Recently, the research focus for SnO_2-based TCOs has shifted to the investigation of tantalum-doped SnO_2 (TTO) thin films because of their superior electrical properties and higher thermal stability in vacuum and air [10–14]. This latter material property is especially critical as these films can find an application as a capacitor electrode in data storage devices, or as a solar-selective transmitter in concentrated solar power plants that require a heat resistance up to 800°C [15, 16]. Because this material demonstrates the potential to be used in a wide range of applications, the necessary tunability to meet the different technical specifications requires the use of scalable low-temperature deposition techniques.

To date, TTO thin films exhibiting the best transport properties were pulsed-laser deposited. $Sn_{0.95}Ta_{0.05}O_2$ films epitaxially grown on rutile TiO_2 (110) single-crystal substrates at 800°C showed a resistivity of 1.1×10^{-4} Ω cm, and $Sn_{0.997}Ta_{0.003}O_2$ films grown at 600°C had the maximum charge carrier mobility for TTO with 130 cm^2 $(Vs)^{-1}$ [11, 17]. The best magnetron-sputtered (MS) TTO films, SnO_2:Ta (2.3 cat. %; cat. stands for cation

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concentration) were grown at 600°C using a radiofrequency (RF) plasma and presented a resistivity value of 5.4×10^{-4} Ω cm [12]. For the growth of SnO_2:Ta at low temperatures between 25°C and 300°C, resistivities from 4×10^{-3} to 1 Ω cm were reported. These studies have been performed so far at typical research lab conditions (Table 1). The sample sizes were, in the majority of cases, 1 cm^2. RF sputtering usually results in very good thin film properties, but has lower deposition rates than direct-current (DC) MS, is technically more challenging and expensive, requires higher voltages, and is sensitive to overheating. Therefore, the present study aims to develop an industry-adaptable thin film growth process for SnO_2:Ta. The transition step from research to industrial scale is crucial for any process implementation in a production line. As magnetron sputtering is a cost-effective, industrially scalable technique, an appropriate pilot-scale setup at Nano4Energy S.L. (Madrid, Spain) was used for the deposition of SnO_2:Ta at low temperature.

X-ray diffraction and Raman spectroscopy were used to identify the phase structure evolution of a series of SnO_x:Ta thin film samples grown as a function of the oxygen flow rate. A distinct resistivity minimum was established at optimized conditions for samples of the composition SnO_2:Ta (0.7 ± 0.05 at.% Ta). These films were then further characterized to build relationships between the deposition temperature, the phase and microstructure, and their electrical and optical properties.

2 | Experimental and theoretical methodology

2.1 | Preparation of SnO_x:Ta Thin Films

SnO_x:Ta thin films were deposited by reactive DC–MS in a pilot-scale sputter chamber of the company Nano4Energy [23]. The system setup consists of two rectangular industrial-size magnetrons (40 × 10 cm, one equipped with a Sn target, the other with a Sn/Ta target 95:5 wt.%) connected to two hiP-V 10 kW power supplies, which can be operated in DC or high-power impulse magnetron sputtering mode. It comprises a motorized twofold rotating system in the middle of the chamber as substrate holder, a shutter for masking the samples and a pair of vertical radiative heaters. Typically, three substrates of fused silica (30 × 30 × 3 mm), Corning glass (same size), and silicon (10 × 10 × 1 mm) were simultaneously mounted onto the sample holder for the depositions. The centrally positioned oxygen inlets over the magnetrons ensure a homogeneous oxygen distribution in the plasma. Following a preoptimization with respect to oxygen flow rate and magnetrons' powers, a systematic series of SnO_x:Ta thin films was grown as a function of the oxygen flow rate at a deposition temperature of 225°C with a working pressure of 1.1 Pa Ar. For the choice of the deposition temperature, two arguments were considered. First, in order to grow crystalline films, the substrate temperature had to be higher than 60°C, which is a realistic temperature for sputter deposition without heating. Secondly, the temperature should represent a kind of mean value between that of the previous low-temperature studies of SnO_2:Ta thin films shown in (Table 1). The oxygen flow rate varied between 12 and 150 sccm, and was controlled using a mass flow controller operated with a GENCOA Speedflo system [24]. After presputtering for 15 min, typical sputter powers were 1000 W for both magnetrons, and the deposition rates were of the order of 1 μm/h. To ensure reproducibility, three depositions were carried under the finally established optimized conditions.

2.2 | Characterization of SnO_x:Ta Thin Films

A combination of X-ray diffraction and Raman spectroscopy was used to analyze the films' phase structure. Grazing incidence XRD measurements were performed on a Rigaku SmartLab diffractometer in the diffraction angle range $15° \leq 2\theta \leq 120°$ with Cu-K_α radiation ($\lambda_{\alpha1} = 1.54059$ Å and $\lambda_{\alpha2} = 1.54441$ Å with the intensity ratio Cu-$K_{\alpha2}$/Cu-$K_{\alpha1}$ = 0.5). The diffractometer was equipped, on the incident side, with a parallel beam mirror, a Soller slit (2.5°), an incident slit (1 mm), and a length-limiting slit (10 mm). The receiving side consisted of a parallel slit analyzer (0.5°) and another Soller slit (2.5°). The incidence angle was 2°, and the measurement was carried out with a step width of 0.05° and a speed of 1.00°/min. The PDF cards 00-004−0673, 04-008−7671, and 00-041−1445 for β-Sn, SnO, and SnO_2, respectively, from the PDF-5+ database were used as references for the identification of structural phases in the studied sample series [25].

The size of the crystallites (D) in the phase-pure SnO_2:Ta thin films was calculated using two methods: the Scherrer equation itself and the Williamson–Hall (WH) plot. The direct use of the Scherrer equation presupposes a strain-free state. With Lorentzian broadening of the diffraction peak, its full width at half maximum (FWHM) β_{exp} is then the sum of size broadening β_{size} and instrumental broadening β_{instr}, or $\beta_{size} = \beta_{exp} - \beta_{instr}$. D is then linked to β_{size} in the Scherrer equation [26–29] The latter is based on the Scherrer constant $K_S = 0.9$ for crystallites without a defined shape, the (average) X-ray wavelength $\lambda = 0.15419$ nm, and the Bragg angle θ

TABLE 1 | Deposition parameters and resistivities of SnO_2:Ta thin films grown at low temperature in previous studies.

θ/°C	Technique	Sample size/cm^2	ρ/Ω cm	Reference
200	RF–MS of SnO_2:Ta target (1 at. % Ta)	?	0.01	Yamada et al. [18]
200	pulsed-MS of SnTa targets	1 × 1	6.4×10^{-3}	Muto et al. [19]
300	RF–MS, ceramic targets	1 × 1	1	Weidner et al. [12, 20]
150	RF–MS, SnO_2 + Ta_2O_5 targets	5 × 5	0.05	Featherstone et al. [21]
25	reactive MS, ceramic SnO_2:Ta target	1 × 1	4×10^{-3}	Mientus et al. [22].
225	reactive DC–MS; Sn + SnTa targets	2 × (3 × 3) + 1 × (1 × 1) in one batch	0.3	This work

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$$D = \frac{K_S \lambda}{\beta_{size} \cos \theta}$$

The Scherrer equation assumes that all crystallites in the film have the same shape and size [29]. Hence, this calculation was done for five diffraction peaks in the diffractogram corresponding to the planes (110), (101), (211), (200), and (002) of rutile SnO_2. The final value determined for each film is the average size calculated from these five peaks, and the error bars were determined from both the standard deviation of the profile fit and the reproducibility of the thin film deposition process.

Since microstrain (ε) is likely to increase the diffraction peaks' FWHM by the amount of β_{strain} and thus reduce the apparent crystallite size, the WH plot, which allows for the determination of the respective contribution of size and microstrain broadening, was applied, too [28, 29]. It is based on a combination of the Scherrer equation with a relation between peak width and microstrain, yielding the equation

$$(\beta_{exp} - \beta_{instr}) \cos \theta = \frac{K_S \lambda}{D} + 4\varepsilon \sin \theta$$

The data points considered are the ones obtained from diffraction peaks that are located in the 2θ range between 20° and 90°, present a high intensity, and yield a small standard deviation when fitted with a Lorentzian profile. When plotting $(\beta_{exp} - \beta_{instr}) \cos \theta$ as a function of $4 \sin \theta$, a linear dependence is expected. The microstrain ε was directly read from the slope of the linear fit, and the crystallite size D was deduced from the intercept [26] However, this method was initially developed for powder samples and not for thin films, so the results were interpreted cautiously [29].

The Raman spectra were measured in a 180° backscattering geometry with a Labram Evolution micro-Raman spectrometer (Horiba) equipped with a 1800/mm holographic grating and a liquid nitrogen-cooled back-illuminated deep-depleted CCD detector. For excitation served the 532 nm radiation of a frequency-doubled Nd:YAG laser, which was focused to a 1 µm spot by a 100-fold magnifying long-working distance objective. The laser power in the spot was 0.75 mW. Typically, Raman spectra from three different sample positions were averaged for spectra fitting analysis employing the software Origin. Fused silica and Corning glass reference spectra were subtracted from the spectra of the TTO samples.

The films with the desired SnO_2:Ta phase structure were further characterized with respect to their element composition and electrical, optical, and electronic properties. The depth-resolved elemental composition of the thin films was analyzed by elastic recoil detection (ERD) using a 43 MeV Cl^{7+} ion beam of a 6 MeV tandem accelerator of the HZDR [30]. In the experimental setup, the angle between the sample normal and the incoming beam was 70°. The scattering angle was 29.5° relative to the direction of the incoming beam. Typically, an area of approx. $2 \times 2\ mm^2$ was irradiated. For the detection of recoils and scattered Cl^{7+} ions, a Bragg ionization chamber was employed. Further details of the ERD measurements are described in previous studies of our group [14, 31]. The spectra of the recoiled species and the scattered Cl ions were fitted simultaneously using the program NDF v9.3g [32]. The experimental accuracy of the ERD analyses is estimated to be ±0.5 at% for Sn and O, and ±0.1 at% for Ta.

For electrical characterization, the sheet resistance of the SnO_2:Ta samples was measured by four-point probe using an AiN CMT-SR2000N four-point probe setup (AiT Co., Ltd.) incorporating a JANDEL four-point probe head with 1 mm probe spacing. The optical transmittance (T) and reflectance (R) spectra of the samples with the best electrical properties were acquired in the wavelength range from 300 to 3000 nm using a spectrometer SolidSpec-3700DUV (Shimadzu). As experimental parameters, a spectral resolution and a step width of 5 nm, and a medium scanning speed were applied. The reflectance was measured under an incidence angle of 5° relative to the surface normal. The absorptance (A) was subsequently calculated based on the relation for energy conservation: $A = 1 - R - T$. A Tauc-plot analysis was used for the estimation of the optical gap of a SnO_2:Ta (0.7 at.% Ta) thin film with the lowest resistivity, following the relation $\alpha h\nu \approx (h\nu - E_{gap})^r$ [33, 34] The expression $(\alpha h\nu)^{1/r}$, depending on the absorption coefficient α and $r = 0.5$ accounting for dipole-allowed optical transitions, was plotted as a function of the photon energy. The linear range of the plot was fitted by a linear function, and the optical bandgap was estimated from its intercept with the energy axis. In addition to UV–Vis–NIR spectroscopy, the same samples were measured by spectroscopic ellipsometry using a variable angle Woollam M2000—XI ellipsometer. Three angles of 55°, 65°, and 75° were applied, and the measurement range was 216 to 1680 nm. The Complete Ease software was used to model thickness, roughness, and optical constants.

X-ray photoemission spectroscopy (XPS) was used to determine the valency of the incorporated Ta. XPS spectra of an optimized SnO_2:Ta sample of this series and of a sample from previous work grown at 570°C in a lab-scale chamber were acquired by the PHI 5600 spectrometer using monochromatic K_α X-ray radiation from an Al anode (200 W) at IFW Dresden. The sampling spot size was 0.8 mm in diameter. To compensate for charging during the measurements, the sample was illuminated with an electron beam of 5 eV energy and an applied current of 18 mA. Survey scans (from −10 to 1300 eV) were collected at an acceptance angle set to 45°, at a pass energy of 29.35 eV and with a step size of 0.4 eV. The measured spectra were calibrated with respect to the C 1s peak at 284.8 eV. To assess the valency of tantalum, the spectra were compared with those of two references, accounting for the different oxidation states of tantalum. The first reference is a 1 mm-thick Ta sheet that was kept for more than 20 years in air, which is thus oxidized and represents Ta_2O_5 (Ta^{5+}). The second one is the same Ta sheet after sputter cleaning with Ar^+ ions, which represents pure Ta (Ta^0). Since the peak of Ta 4f overlaps with that of Sn 4d, the spectra of Ta 4d were used to evaluate the valency of Ta.

3 | Results and Discussion

3.1 | Phase Structure Evolution as a Function of the Oxygen Flow Rate during Low-Temperature Growth of SnO_x:Ta Thin Films

The X-ray diffractograms of the studied SnO_x:Ta films reveal a systematic evolution of the phase structure as a function of the O_2 flow rate (Figure 1). The film grown with the lowest O_2 flow rate of 12 sccm has the highest number of diffraction peaks compared to the other films, which is a first indication

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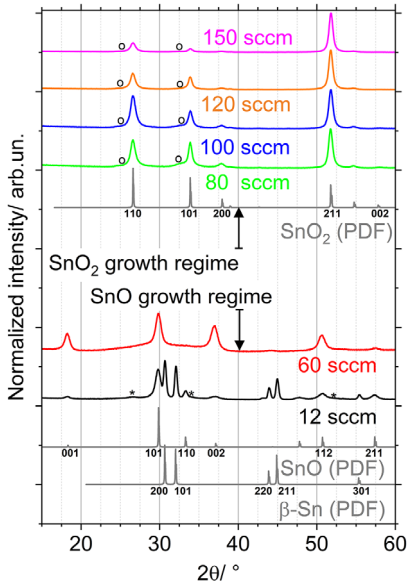


FIGURE 1 | X-ray diffractograms of SnO_x:Ta thin films grown at 225°C as a function of the oxygen flow rate during deposition. The powder diffractograms of β-Sn, SnO, and SnO_2 are plotted as a reference for comparison. The diffractograms are normalized and shifted along the y-axis for clarity. * indicates a weak diffraction peak of SnO_2; o indicates a shoulder-like diffraction peak of unknown origin.

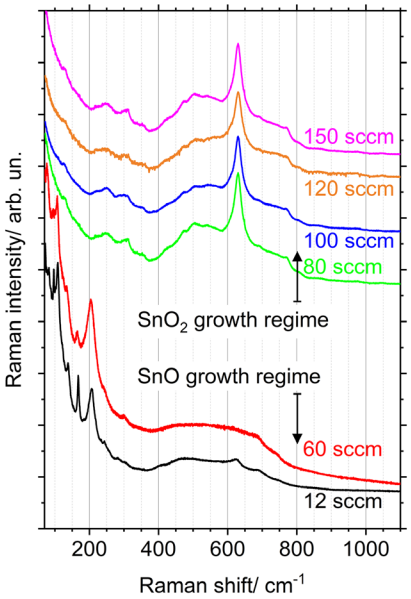


FIGURE 2 | Raman spectra of SnO_x:Ta thin films grown at 225°C as a function of the oxygen flow rate during deposition. The spectra are shifted along the y-axis for clarity. An expanded low-energy spectrum of the film grown with 12 sccm O_2 with an assignment of the Raman lines to structural phases is given in (SI 1).

of a mixture of different structural phases. The identification of SnO and β-Sn follows from the match between the observed and the reference powder diffractograms (Figure 1). The SnO diffraction peaks are broader than those of β-Sn. This is an indication for a smaller crystallite size of SnO than of β-Sn, which is attributed to the higher diffusivity in metals and to the higher melting temperature of SnO. In addition to these, three shoulder-like or weak diffraction peaks at $2\theta = 26.5°$, $2\theta = 34.3°$, and $2\theta = 52.1°$ can be assigned to the lattice planes (110), (101), and (211) of SnO_2, respectively, which represents a minority phase in this sample. The film grown at 60 sccm is characterized by the dominance of the characteristic diffraction peaks of SnO. The absence of β-Sn is consistent with the higher oxygen flow rate during deposition. Moreover, this film exhibits a higher background in the 2θ range between 20° and 40°. Since this background is localized on a rather narrow range of 2θ, it cannot be attributed to inelastic scattering. It could rather be an indicator for the coexistence of crystalline SnO and an amorphous volume fraction, whose diffraction peaks are extremely broadened and blend together to form this background.

For films grown at higher O_2 flow rates from 80 to 150 sccm, the reference diffractogram of rutile SnO_2 matches perfectly. There are no additional diffraction peaks that could be attributed to other phases of the Sn–O system, or to a phase containing tantalum (like Ta_2O_5, for example) [25]. Two weak, shoulder-like peaks at $24.4° < 2\theta < 24.7°$ and $31.3° < 2\theta < 32.8°$, respectively, could not be assigned to any known compound of the Sn–O–Ta system. Their appearance might be the result of lattice strain [35], or they originate from the presence of an unidentified minority phase [21].

The Raman spectra of the SnO_x:Ta films confirm the distinctly different phase structure between the films grown with O_2 flow rates up to 60 sccm and from 80 sccm on upwards (Figure 2). The spectrum of the film grown at the lowest oxygen flow rate of 12 sccm is characterized by eight significant lines at wave numbers below 250 cm^{-1} (SI 1). The highest intensity is observed for the lines at 108 and 205 cm^{-1}, which are assigned to the B_{1g} (or E_g: the assignment is controversially discussed in the literature) and A_{1g} modes of SnO (Table 2) [36–40]. A second group comprising four lines with medium intensity is assigned to monoclinic Sn_3O_4 (Table 2) [37, 38, 40]. These lines have a much narrower width than those of SnO. At 123 cm^{-1}, a weak but distinct shoulder is found at the high-energy tail of the B_{1g} (E_g) mode of SnO. This line coincides reasonably well with the E_g Raman line of metallic β-Sn at 127 cm^{-1} [41, 42], and has no good counterpart in the reference spectra of SnO and Sn_3O_4. While the presence of SnO, Sn_3O_4, and β-Sn in the film grown with 12 sccm O_2 follows from the low-energy Raman spectra, the presence of SnO_2:Ta is evident for its fingerprint line at 625 cm^{-1}. It is superimposed on the broad feature covering the range from approx. 380 to 800 cm^{-1}.

While the observation of SnO, β-Sn, and SnO_2 is in full agreement with the XRD analysis, the detection of Sn_3O_4 points to a selective enhancement effect for this phase. In fact, resonance Raman enhancement of Sn_3O_4 for laser radiation of 532 nm can be expected, since the laser energy of 2.35 eV matches the bandgap energy of 2.4 eV of monoclinic Sn_3O_4 [43, 44]. It enables the detection of a minority fraction of this phase in the samples grown at low O_2 flow rates, which are predominantly comprised of SnO and β-Sn, and another minority phase of SnO_2:Ta.

The strongest Raman features of the low O_2 flow rate growth regime of SnO_x:Ta are also present in the spectrum of the film grown with 60 sccm. Noticeable changes compared to the spectrum of the 12 sccm sample are the broader widths of the majority of SnO and Sn_3O_4 lines, the absence of the shoulder of β-Sn, and the absence of the SnO_2 fingerprint line at 625 cm^{-1} (Table 2).

The increase of the O_2 flow rate from 60 sccm to 80 sccm results in rutile SnO_2-dominated Raman characteristics, which remain

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TABLE 2 | Selected Raman line parameters and their assignment to structural phases of SnO_x:Ta thin films grown as a function of the O_2 flow rate.

Structural phase	12 (60) sccm O_2 flow rate			100 sccm O_2 flow rate			Reference values (assignment)	Ref.'s
	ν/cm^{-1}	FWHM/cm^{-1}	I_{rel}/%	ν/cm^{-1}	FWHM/cm^{-1}	I_{rel}/%		
SnO	108 (107)	11 (10)	30 (13)	—	—	—	105 … 115 (B_{1g} or E_g)	[36–40]
	206 (204)	21 (32)	39 (54)	—	—	—	204 … 211 (A_{1g})	[36–40]
Sn_3O_4	84 (78)	4 (12)	2 (9)	—	—	—	73	[37]
	97 (97)	3 (13)	4 (8)	—	—	—	90	[37, 38]
	138 (136)	7 (11)	6 (5)	—	—	—	130 … 140	[37, 38, 40]
	168 (165)	4 (7)	6 (2)	—	—	—	170 … 173	[37, 38, 40]
β-Sn	123 (-)	9 (-)		—	—	—	127	[41, 42]
Sn_3O_4 or SnO_2:Ta	246 (245)	21 (13)	6 (2)	252	35	n.a.	238 … 244	[37, 38]
SnO_2:Ta	625 (-)	n.a.	n.a.	611	55	17	A_{1g} (low), (L_4)	[14, 45]
	—	—	—	631	22	17	A_{1g} (high), (L_5)	[14, 45]
	—	—	—	772	23	3	B_{2g} (L_8)	[14, 45]

unchanged when the flow rate is stepwise increased to 150 sccm. The most intense line is the A_{1g} mode, which comprises a shoulder at 611 cm^{-1} and the main peak at 631 cm^{-1} (SI 2, SI 3). The two A_{1g}-derived components were previously denoted by L_4 and L_5 (L for lattice modes in comparison to D for defect modes) [14, 45]. In view of the absence of the D_1 and D_2 defect lines in the present sample series and of the higher number of lattice modes resolved for the samples of the present study in comparison to the former two [14, 45], the notation A_{1g} (low) and A_{1g} (high) is preferred in this work. In addition to the A_{1g} line, the symmetry-allowed B_{2g}-derived line of SnO_2 is safely identified at 772 cm^{-1} (SI 2, SI 3). The usability of the symmetry species of rutile SnO_2 for SnO_2:Ta is in principle not proven. The substitution of one Sn atom by Ta reduces the local bonding symmetry, and the unit cell for the description of the translation symmetry of the Ta-doped SnO_2 crystal is significantly larger than that of rutile SnO_2 with two formulae units in it. Thus, all modes split in several components. In SnO_2, the Sn atom occupies a site of the symmetry 2/m 2/m 2/m. For the special case, the Ta is located in the center of the new unit cell, and assuming a homogeneous distribution of all dopant atoms, the resulting lattice could in fact have orthorhombic symmetry. Then, the symmetry species notation A_{1g} and B_{2g} would still apply, along with an increased number of such modes which have their origin in the original A_{1g} and B_{2g} of the ideal host lattice.

An important question is whether the cosputtered Ta is incorporated into the lattices of the identified phases of the Sn–O system. The substitution of Sn by Ta distorts the lattice structure, reduces the lattice symmetry, and is expected to weaken the symmetry selection rules of the host crystals. This is apparently not the case for the majority phases formed at low O_2 flow rates. Neither the number of lines of SnO and β-Sn nor their frequencies are different from those of the pure phases reported in the literature (Table 2). For the minority phase Sn_3O_4, the situation is not fully clear. The number of strong lines fits with that of the reference data, but some of the frequencies differ nonuniformly by up to ±10 cm^{-1}. However, due to its small volume fraction, which follows from its missing diffraction peaks, its effect on the Ta distribution should be negligible. Therefore, it can be concluded that the Ta atoms in the samples of the two lowest O_2 flow rates are located at interstitials or grain boundaries. In contrast to that, the observed splitting of the A_{1g} mode of SnO_2, located at (633 ± 1 cm^{-1}) in SnO_2 single crystals [46, 47], into two lines at 631 (A_{1g}—high) and 611 cm^{-1} (A_{1g}—low) and the appearance of numerous symmetry-forbidden lines in the full spectral range indicate the substitution of Sn by Ta in the SnO_2 lattice for samples grown at O_2 flow rates from 80 to 150 sccm. Downshift and splitting of the A_{1g} line were also found in our previous DFT calculations, wherein the spectra of a SnO_2 single crystal were compared with those of SnO_2:Ta (1.05 at.% Ta) [14]. The smaller downshift of the two A_{1g} components compared to the previously calculated (593 cm^{-1} and 601 cm^{-1}) and experimental (625 and 605 cm^{-1}) spectra is probably linked to the lower concentration of tantalum in the films. The SnO_2:Ta samples of the present study, in fact, show a significantly lower Ta concentration of (0.70 ± 0.05) at.% compared to those of our previous reports with 1.25 at. % Ta (Table 3).

In conclusion, complementary XRD and Raman spectroscopic data of a series of SnO_x:Ta thin films revealed the formation of a mixed majority phase structure comprising crystalline SnO and β-Sn at low, and of Ta-doped SnO_2 at high oxygen flow rates during DC–MS deposition at 225°C in a pilot-scale deposition setup. In the following section, we focus on the properties of the four samples with rutile SnO_2 phase structure, a Ta concentration of (0.70 ± 0.05) at.%, and a slight excess of oxygen compared to the ideal oxygen/metal stoichiometry of 2/1 (Table 3).

3.2 | Electrical, Optical, Structural, and Electronic Characterization of SnO_2:Ta (0.7 at.%Ta) Thin Films Grown at 225°C

The film's optical and electrical characteristics are key properties for its applications. The evolution of the sheet resistance in the SnO_2:Ta samples as a function of the O_2 flow rate during deposition exhibits a distinct minimum for a flow rate of 100 sccm (Figure 3a). For this sample, presenting the lowest sheet resistance and a thickness of 2 µm, a specific resistivity of 0.3 Ω cm was estimated.

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TABLE 3 | Element composition of the SnO_2:Ta thin films as obtained by ERD. For the discussion, we neglect the amounts of hydrogen and carbon and use the concentrations of Ta, Sn, and O normalized to the sum of their concentration, as, e.g., $c_{Ta,\,norm} = c_{Ta}/\,(c_{Ta}+c_{Sn}+c_O)$.

O_2 flow rate	c_{Ta} [$c_{Ta,\ norm}$]	c_{Sn} [$c_{Sn,\,norm}$]	c_O [$c_{O,\,norm}$]	c_H	c_C	$c_O/(c_{Sn}+c_{Ta})$
sccm	at.%	at.%	at.%	at.%	at.%	1
80	0.74 [0.76]	30.4 [31.2]	66.2 [68.0]	2.7	0	2.13
100	0.74 [0.76]	29.5 [30.3]	67.2 [69.0]	2.3	0.2	2.22
120	0.68 [0.70]	29.8 [30.7]	66.5 [68.6]	3.0	0	2.18
150	0.75 [0.78]	29.8 [30.5]	67.0 [68.7]	2.2	0.2	2.19

This value lies in the upper range of low-temperature-deposited SnO_2:Ta films from research lab studies (Table 1). Its transmittance of 80.6 % in the visible range from 380 to 780 nm (Figure 3b) is equivalent to the values for low-temperature-grown SnO_2:Ta in the literature [18–22]. In contrast to these studies, the range of $T > 80$ % is extended up to 2500 nm. This finding corresponds to a reflectivity of less than 20 % up to this wavelength. The reflectance edge in the NIR range, which is characteristic for charge carrier concentrations of $\geq 2 \times 10^{20}$ cm^{-3} [16], is missing.

Moreover, the linear relationship between the order of the transmittance maxima and their wave number (Figure 3c) implies an almost constant refractive index in the wavelength range from 520 to 2300 nm. This is consistent with ellipsometric modeling for the experimentally accessible transparent range, which gives a slight increase of the mean refractive index from approx. 1.70 at 1680 nm to 1.75 at 500 nm (SI 4). This contrasts to highly conductive SnO_2:Ta, whose refractive index decreased from 2.1 at 300 nm to 0.5 at 2500 nm [16]. The Tauc plot analysis of optimized SnO_2:Ta (0.70 at.% Ta) reveals a linear relationship between the photon energy and $(\alpha h\nu)^2$, for the energy range from 3.65 to 3.87 eV (Figure 3d). This relationship is established for a dipole-allowed, direct optical transition of crystalline semiconductors. The absorption coefficient was calculated by dividing the absorptance A by the film thickness d of 2108 nm

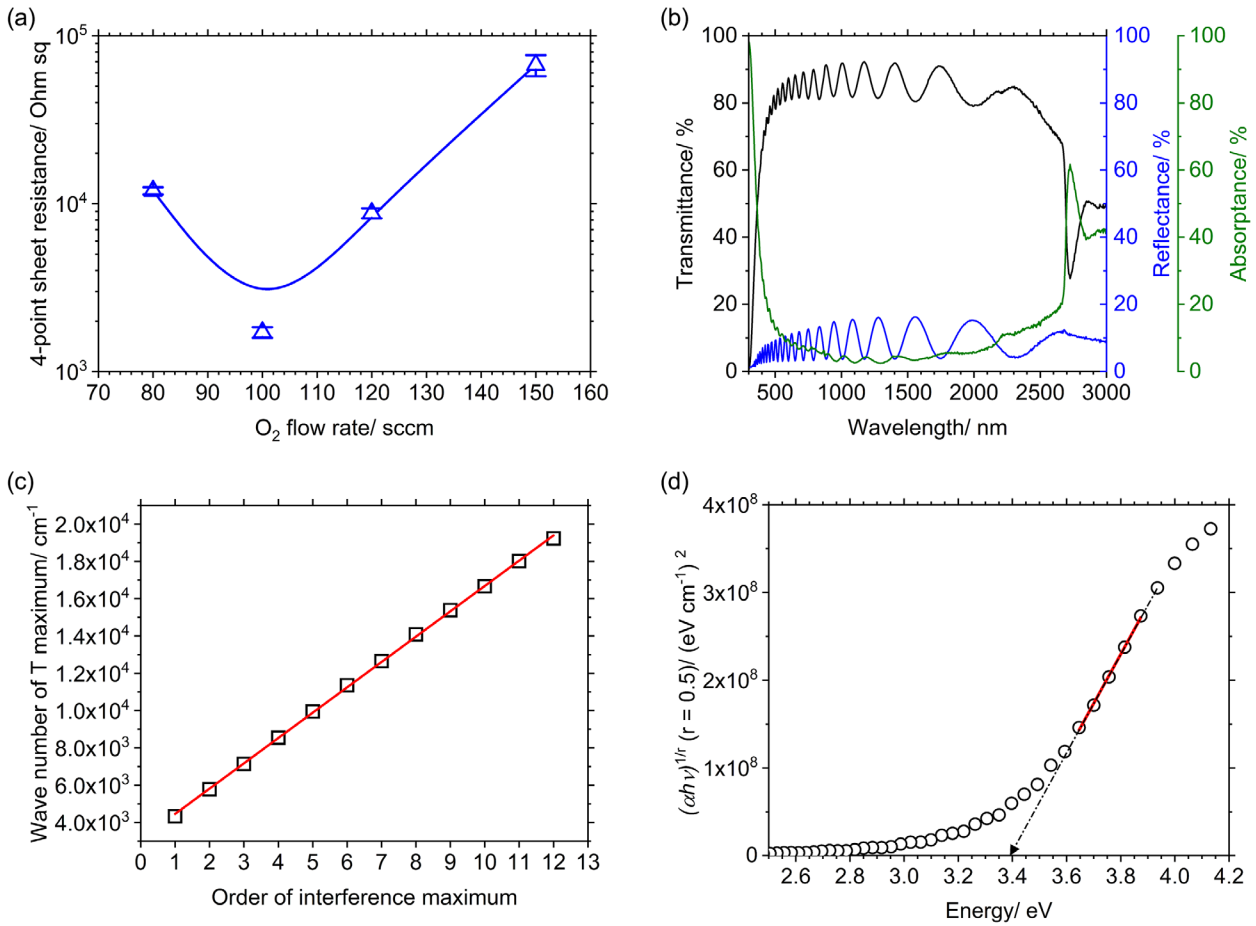


FIGURE 3 | (a) Four-point sheet resistance of SnO_2:Ta thin films grown at 225°C as a function of the oxygen flow rate during deposition. The data represent the mean values and the standard deviations of the data for the films grown on fused silica and Corning. (The data points are connected by a B-spline function as guide for the eye.), (b)–(d) Optical spectra, linear relationship of transmittance (T) maximum wave number and interference maximum, and Tauc plot analysis of an optimized SnO_2:Ta (0.7 at.% Ta) thin film grown at 100 sccm O_2 flow rate on fused silica.

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obtained by ellipsometric modeling, according to $\alpha = A/d$ (SI 4). The intercept of the linear curve with the abscissa, 3.39 ± 0.10 eV, corresponds to the optical bandgap of the studied film SnO_2:Ta (0.7 at.% Ta). Such a small bandgap for the principal valence band (VB) to conduction band (CB) transition in SnO_2 can only be explained by a significant distortion of the electronic band structure, e.g., by broadening of the VB and CB density of states. Moreover, it precludes a significant occupation of the CB by free charge carriers, which would increase the interband transition energy due to the so-called Burstein–Moss effect. In principle, point defects in the bandgap could also lead to a smaller bandgap. On the other hand, they would hardly be expected to affect the principal interband transition because of their low concentration and low transition dipole moment. However, such defects could act as traps for free charge carriers and reduce their concentration. Such a hypothesis was previously formulated by Weidner et al. based on optical data, which indicated a more than three times higher defect concentration in the bandgap of SnO_2:Ta thin films grown at 300°C than in those grown at 600°C [20]. The nature of such point defects would still have to be clarified, since the previously found O_i-type and V_{Sn}-type defects [45] can be excluded because of their absence in the Raman spectra excited with their resonance wavelengths of 532 and 405 nm.

Possible effects of the microstructure on charge carrier formation and activation are addressed by applying the Scherrer equation and a WH analysis to the XRD data of the SnO_2:Ta (0.7 at. % Ta) thin films grown at O_2 flow rates from 80 to 150 sccm. The Scherrer equation gave crystallite sizes from 15.6 to 16.3 nm with standard deviations of (+2.5/−2.4) nm (Figure 4a). Hence, the crystallite size of the four SnO_2:Ta films is independent of the oxygen flow rate. The WH plots indicate some amount of microstrain in these samples, which becomes visible by a positive slope in the linear fits of the data (Figure 4b). Since the WH analysis was developed for powder samples, and not for thin films, and especially since the measurements were carried out in grazing incidence geometry, the results are only used as a first estimate of the presence of microstrain in the film [29]. For instance, a microstrain of 0.28% was estimated for the sample with the lowest resistivity grown at 100 sccm O_2 (Figure 4b). This yields a slightly larger crystallite size of 18.3 nm for this specific sample compared to 16.0 nm obtained by the Scherrer equation. In comparison to our previous SnO_2: Ta (1.25 at. % Ta) thin films grown at 570°C, which had a mean crystallite size of 58 nm, both the Scherrer equation and the WH analysis reveal a significantly smaller crystallite size (by a factor of 3 to 3.5 in one dimension) for the low-temperature-grown samples of the present study [14]. The smaller size can be linked to the lower substrate temperature during deposition, which is responsible for slower surface diffusion of the species, leading to smaller crystallites. To some extent, the smaller crystallite size could explain the observed electrical properties in the sense that the high grain boundary density related to small crystallites leads to a higher scattering rate of the free charge carriers and ultimately to a lower conductivity. The ellipsometric modeling indicates a significant roughness of the order of 75 nm, and a graded refractive index for the spectral range in which the samples are transparent, both supporting the conclusion of a strongly distorted microstructure (SI 4). In addition, in a first qualitative estimate, the films exhibit microstrain. This accounts for a variation of the interplanar spacing occurring in the vicinity of defects, such as dislocations or stacking faults, which are also unfavorable for efficient conductivity due to scattering of free charge carriers. Overall, the microstructure of the studied films does not support optimal transport properties.

To check whether the Ta dopant exists in a state of Ta^{4+} in the SnO_2 lattice, which would prevent the formation of charge carrier and their excitation into the CB, Ta-4d XPS spectra of the current SnO_2:Ta (0.7 at.% Ta) films were compared to those of SnO_2:Ta (1.25 at.% Ta) of our previous studies and of two reference samples representing the oxidation states Ta^{4+} and Ta^{5+}. Obviously, there is no indication of Ta^{4+} in the spectra of the samples because it would appear at a lower energy (Figure 5). Instead, the peak energies of the SnO_2:Ta (0.7% at. Ta) film are the same as the ones of the conductive film (SnO_2:Ta (1.25% at. Ta)) and the oxidized Ta sheet. It can be deduced that both films present Ta species in the state Ta^{5+}. Consequently, a valency of Ta^{4+} cannot be the reason for the relatively high resistivity exhibited by the films in the present series, nor can it be the reason for the low bandgap energy.

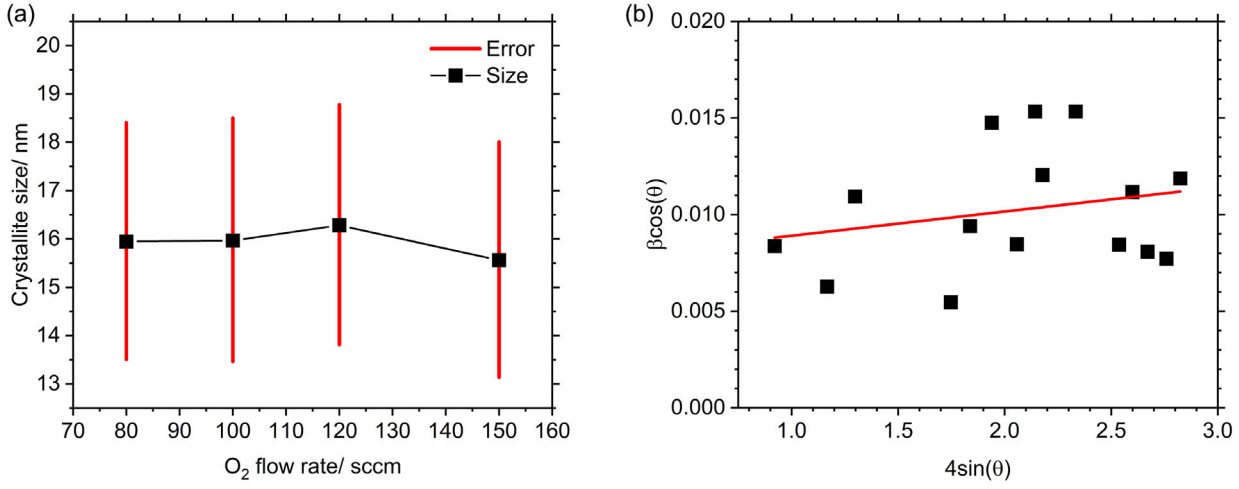


FIGURE 4 | (a) Crystallite size of SnO_2:Ta films as a function of O_2 flow rate obtained by the Scherrer equation. (b) WH plot for the SnO_2:Ta (0.7 at.% Ta) film grown with 100 sccm O_2 flow rate. The black line in (a) is a guide for the eye; the red line in (b) a linear fit to the data. WH = Williamson–Hall.

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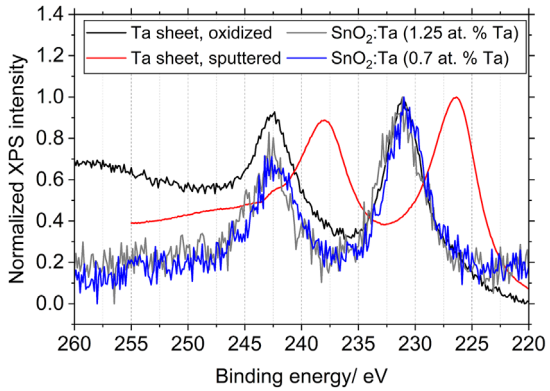


FIGURE 5 | Ta 4d core level spectra of the optimized SnO_2:Ta (0.7 at.% Ta) thin film in comparison to the XPS spectra of a freshly prepared metallic Ta film, an oxidized Ta film, and a conductive SnO_2:Ta (1.25 at.% Ta) thin film. The spectra are normalized relative to the corresponding maximum intensity for better comparison. XPS = X-ray photoemission spectroscopy.

4 | Conclusions

The formation of the TCO SnO_2:Ta was studied at a low temperature of 225°C in a pilot-scale PVD chamber using reactive direct-current magnetron sputtering from 40 × 10 cm metallic targets. The process is suitable for scaling up to real-world industry conditions. A systematic phase structure evolution was observed consistently by X-ray diffraction and Raman spectroscopy across a series of samples deposited with increasing oxygen flow rates. At low oxygen flows, SnO and β-Sn were formed as the majority phases, while Sn_3O_4 and SnO_2 as minority ones. At high oxygen flow rates, single-phase rutile SnO_2 was observed to grow. The Raman spectra revealed the substitutional incorporation of the dopant in the rutile structure, resulting in transparent conductive SnO_2:Ta (0.7 at.% Ta). No Ta incorporation was found in SnO.

Among the four SnO_2:Ta (0.7 at.% Ta) samples grown in the high O_2 flow rate regime, a distinct sheet resistance minimum was found for the flow rate of 100 sccm O_2. Further characterization of the optimized sample yielded a very good transmittance of 80.6 % in the visible spectral range, but a missing reflectivity edge in the NIR spectral range, a relatively high specific resistivity of 0.3 Ω cm, and an unusually small optical bandgap of 3.39 ± 0.10 eV. Data of the element analysis and of the XRD, and XPS spectra were used to investigate the reasons for a) the overall relatively high resistivity and b) the evolution of it as a function of the oxygen flow rate in the SnO_2:Ta sample series. Since the film's element composition was identical (Table 3), it cannot explain any of those observations. Scherrer equation and WH analysis revealed a small mean crystallite size of 16–19 nm and the presence of microstrain for all SnO_2:Ta samples. Both could reduce the conductivity and be at least in part responsible for the relatively high resistivity due to scattering of charge carriers at grain boundaries and at lattice faults. A low oxidation state of Ta^{4+}, which could have prevented the charge carrier formation at all, was excluded by comparison with XPS spectra of highly conductive SnO_2:Ta (1.25 at.%Ta) and suitable references for Ta^{4+} and Ta^{5+}. Point defects in the bandgap formed due to the low growth temperature could also be responsible for the comparably high resistivity [20]. Presumably their concentration reaches a minimum for the O_2 flow rate of 100 sccm, resulting in the observed distinct resistivity minimum. Moreover, we argue a broadening of the valence and CB density of states as a result of the microstructure, which might explain the small bandgap. While reactive DC–MS represents an appropriate, scalable, low-temperature PVD technique for the deposition of transparent conductive Ta-doped SnO_2 thin films, improvements of the process are necessary. Promising modifications would include the application of a higher sputter power density and in the use of pulsed techniques in order to achieve higher growth rates and a lower defect density.

Author Contributions

Matthias Krause: Supervision, conceptualization, methodology, investigation, formal analysis, data curation, writing – original draft, writing – review & editing, corresponding author. **Cecilia Bauden**: Investigation, formal analysis, data curation, writing – original draft, writing – review & editing. **Justus Haag**: Investigation, formal analysis, data curation, writing – review & editing. **Alvaro Mendez**: Investigation, formal analysis, data curation, writing – review & editing. **Frans Munnik**: Investigation, formal analysis, data curation, writing – review & editing. **Fabian Ganss**: Investigation, formal analysis, data curation, writing – review & editing. **Ambiörn Wennberg**: Conceptualization, methodology, writing – review & editing. **Ivan Fernandez**: Conceptualization, methodology, writing – review & editing. **Ivan Kaban**: Investigation, formal analysis, data curation, writing – review & editing. **Harish Parala**: Supervision, writing – review & editing. **Anjana Devi**: Supervision, writing – review & editing.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Supporting Information

Additional SI can be found online in the Supporting Information section. **Supporting Fig. S1**: Low-energy raw data and peak fit analysis for the Raman spectrum of the SnO_x:Ta thin film grown with 12 sccm O_2 flow rate. **Supporting Fig. S2**: High-energy Raman spectrum of the SnO_2:Ta thin film grown with 100 sccm O_2 flow rate measured with 532 nm laser excitation and peak fit analysis. **Supporting Fig. S3**: High-energy Raman spectrum of the SnO_2:Ta thin film grown with 100 sccm O_2 flow rate measured with 405 nm laser excitation and peak fit analysis. **Supporting Fig. S4**: Ellipsometric data, fit model, and model parameter for the transparent range of the SnO2:Ta thin from 500 nm to1680 nm film grown with 100 sccm O_2 flow rate: a) Parameter Psi, b) Parameter Delta, c) optical constants.

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